

Comparison of Physically simulated Monte-Carlo Simulation and Numerical Simulated Buffon's Needle Experiment with induced Uniform and Gaussian Bias

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Abstract

Buffon's Needle experiment is a *physical* Monte Carlo simulation that obtains π by tossing toothpicks with two key variables: randomness of the scattering of the toothpick centers, and the angle of orientation of the toothpicks which induces π . In this paper, we observe how this demonstration converges to π through the Law of Large Numbers (LLN) and Central Limit Theorem (CLT) as N increases. Upon comparing these results with numerically simulated models (uniform bias and gaussian bias) and calculating the value of π from experimentation, we observe that there is percent error of less than 1.2 and comparing with simulations, there is a strong agreement between both models converging to π . This shows that Buffon's Needle Experiment is a strong method in statistically estimating π .

Keywords: Monte Carlo methods, Buffon's Needle, geometric probability, Law of Large Numbers, physical simulation

1 Introduction

Monte Carlo simulations constitute a broad class of numerical techniques that estimate quantities using repeated random sampling. Formally, they approximate integrals, expectations, or probability distributions by drawing samples from a stochastic process and computing ensemble averages. Their validity follows from the Law of Large Numbers (LLN) and the Central Limit Theorem (CLT), which guarantee convergence to the true expectation as the number of samples increases [1].

A *physical* Monte Carlo simulation is a realization of these same principles using randomized physical processes instead of algorithmically generated pseudorandom numbers. Rather than computing samples numerically, one uses a physical system whose configuration space is effectively random, such as thermal motion, scattering processes, or—as in the present study—the random orientation and placement of falling toothpicks. From a statistical perspective, each drop constitutes an independent Bernoulli trial, forming an empirical estimate of a theoretically known probability [2].

The estimation of π via dropping toothpicks arises from Buffon's Needle Problem. Consider a needle (or toothpick) of length c dropped on a plane ruled with parallel lines separated by distance a . The probability that the needle intersects a line is [3]

$$P = \frac{2c}{\pi a}, \quad c \leq a. \quad (1)$$

This expression results from integrating over the joint distribution of the needle's center position and orientation angle θ . Solving for π yields

$$\pi \approx \frac{2c}{aP_{\text{exp}}}, \quad (2)$$

where P_{exp} is the experimentally measured crossing probability, i.e., M/N , where N is the number of dropped toothpicks and M is the number of intersection. By the Law of Large Numbers, the empirical probability converges to its theoretical value as the number of drops increases, improving the accuracy of the estimate. The statistical uncertainty scales as $1/\sqrt{N}$, where N is the number of trials.

In practice, physically dropping toothpicks does not produce a perfectly uniform distribution of positions and orientations. Human motion typically results in a two-dimensional Gaussian-like positional distribution rather than a uniform one, and slight orientation biases may occur. However, the crossing

probability depends only on the conditional probability of intersection given a position and angle. Provided that the orientation distribution remains approximately uniform and is not strongly correlated with landing position, the resulting estimate of π is only minimally affected. Deviations from ideal randomness therefore introduce only small corrections and do not significantly alter the experimental outcome [4].

The goal of this paper is to experimentally verify that randomly dropped toothpicks yield an empirical crossing probability consistent with Buffon's theoretical prediction, thereby producing a statistically meaningful estimate of the constant π . And finally, identify (if any) key differences between a uniform bias and a gaussian induced bias in the toothpick center scattering.

2 Methodology

2.1 Required Materials and Experimental Setup

Several boxes of wooden toothpicks, a meter stick, and large sheet of chart paper was used in this experiment. The wooden toothpicks had a length c of 65 mm. There were parallel lines drawn on the chart paper each separated by distance a of 130 mm so that the length of the toothpicks were smaller than the separation distance.

2.2 Procedure

The experiment was conducted by dropping a certain number of toothpicks from a height of 1 m onto the chart paper below. Then, the amount of toothpicks that intersected with the parallel lines of the chart paper was counted and recorded. This setup was repeated for three trials and for each toothpick quantity. The number of toothpicks started with 10 and increased by an increment of 10 until there were a total of 150 toothpicks that were tested. For each quantity, the three recorded values were averaged, and this average was used in the calculation of π according to the formula.

In addition, numerical simulations were made to compare the experimental results with the theoretical expectations. Specifically, two modes of simulations were tested, one with a normal distribution and the other with a uniform distribution. Then their results were also graphed. In this case, an average of 10 trials per increment was used instead.

$$\pi_{\text{exp}} = \frac{2cN}{aM} \quad (3)$$

where N is the total number of toothpicks dropped and M is the corresponding number of intersections.

The following data was then compared with numerically simulated models for a uniform bias, and then a gaussian bias in the toothpick center scattering.

3 Results and Discussion

Figure 1 illustrates the behavior of the experimentally obtained estimates of π using physically dropped toothpicks calculated by equation (3). At low sample sizes ($N \leq 20$), the values fluctuate significantly due to the large statistical variance inherent in small-sample Monte Carlo trials. This is reflected in the first two data points, which both yield a percent error of 36.4%. Such deviations are expected, as the variance of a Bernoulli estimator scales as $P(1 - P)/N$, meaning that the relative noise is dominant at small N . As the number of dropped toothpicks increases, the results begin to stabilize and trend toward the theoretical value of π . The minimum experimental percent error obtained was 1.21% at $N = 120$, which demonstrates that even with moderate sample sizes, physical Monte Carlo methods can reasonably approximate the expected theoretical limit.

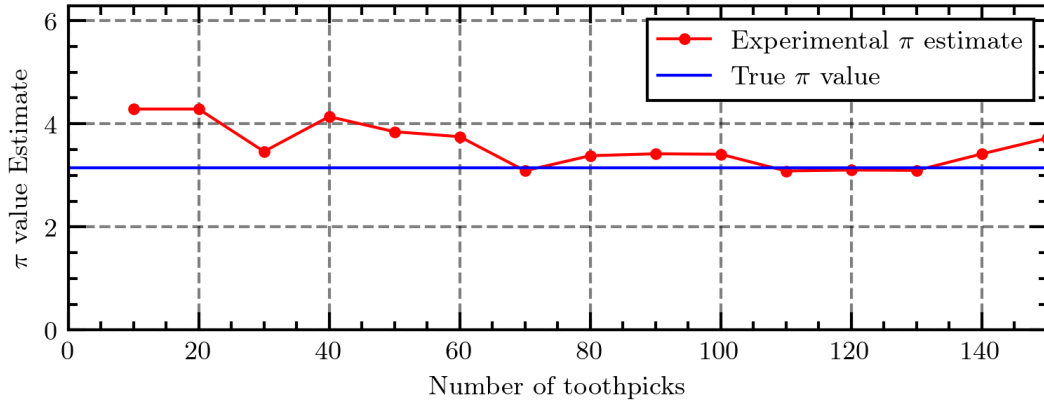


Figure 1: Experimental results of calculating the estimated value of π , at 3 trials averaged at each point. The value closely approaches π , but not quite. However, it can be assumed that at high enough N , the values will converge.

Table 1 summarizes the numerical values associated with Figure 1, displaying the characteristic non-monotonic convergence behavior typical of stochastic sampling. It shows that at higher N , there is also more increments, which probabilistically converges to π .

Table 1: Pi Estimation Results

Increments	Average Hits	π Estimate	Percent Error (%)
10	2.333	4.286	36.42
20	4.667	4.286	36.42
30	8.667	3.462	10.18
40	9.667	4.138	31.71
50	13.000	3.846	22.43
60	16.000	3.750	19.37
70	22.667	3.088	1.70
80	23.667	3.380	7.60
90	26.333	3.418	8.79
100	29.333	3.409	8.51
110	35.667	3.084	1.83
120	38.667	3.103	1.21
130	42.000	3.095	1.48
140	41.000	3.415	8.69
150	40.333	3.719	18.38

Figure 2 shows the results of the numerical Monte Carlo simulation, averaged over ten trials per data point. Unlike the physical experiment, which is very time consuming and as such can be performed with less data points in time, the simulation enforces ideal sampling conditions, such as uniform angle distributions and consistent Gaussian spatial distributions. As a consequence, the simulated data converges more smoothly and achieves percent errors below 1% at comparatively lower N . Also, the results do not vary regardless if one uses a uniform or gaussian distribution for the toothpicks. This contrast shows two key results : (1) the geometric probability model underlying Buffon's problem is indeed correct and robust, and (2) physical sources of bias including imperfect randomization of drop angles and nonuniform landing distributions, contribute only secondary deviations and do not fundamentally alter the convergence trend. The general similarity between the physical and simulated curves confirms that the experiment behaves as theoretically predicted, validating the applicability of Monte Carlo principles and the Law of Large Numbers to this physical system.

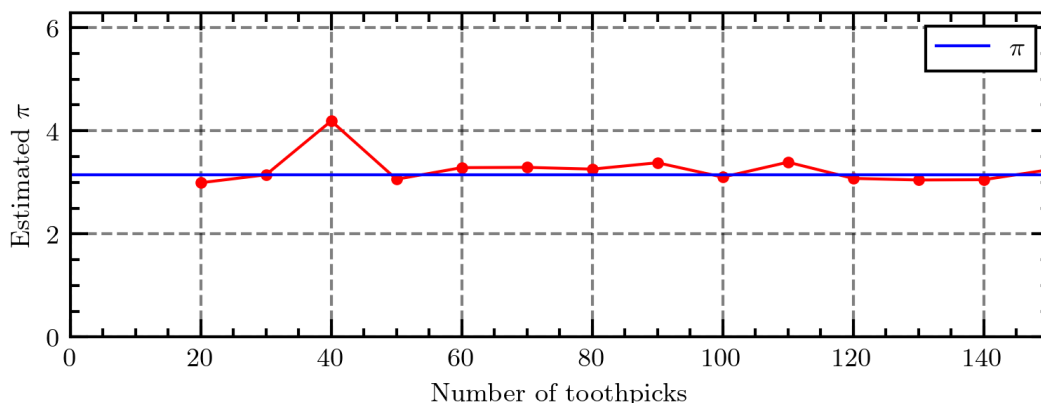


Figure 2: Numerical Monte Carlo simulations of the same experiment. This graph is showing the Gaussian bias and an averaged. Note that the estimated value more converges faster and is more accurate compared to the physical results

4 Conclusion

The experimental and simulated results collectively demonstrate that Buffon's Needle experiment provides a viable physical Monte Carlo method for estimating the value of π . Despite the inherent imperfections of physical randomization, the experimental data clearly follows the theoretical convergence trend predicted by geometric probability. While the early data points exhibit large fluctuations due to small sample sizes, the estimates steadily improve as N increases, eventually achieving percent errors close to 1%. This behavior is fully consistent with the statistical foundations of Monte Carlo techniques, with variance reduction through repeated independent trials.

Furthermore, the comparison with the numerical simulation highlights the strong agreement between theoretical predictions and idealized computational behavior. The simulation converges rapidly and with high precision, serving as a reference for the expected outcome under perfect randomness. The physical experiment, though noisier, ultimately validates these predictions and demonstrates that real-world random processes are sufficiently unbiased to approximate geometric probabilities accurately. Thus, the study confirms that randomly dropped toothpicks indeed approach a statistically meaningful estimate of π .

References

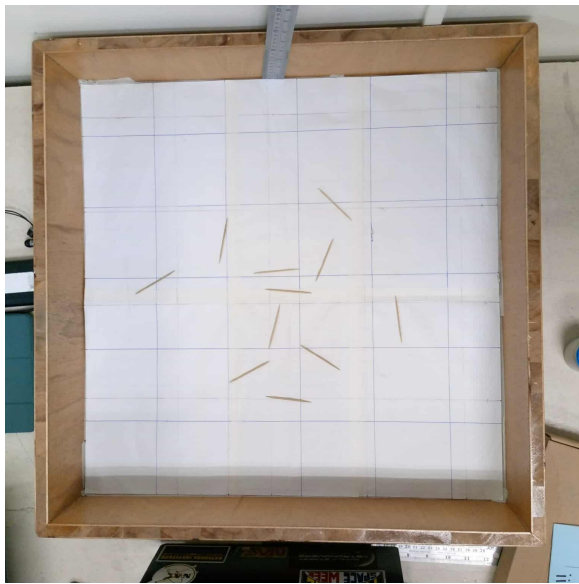
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Appendices

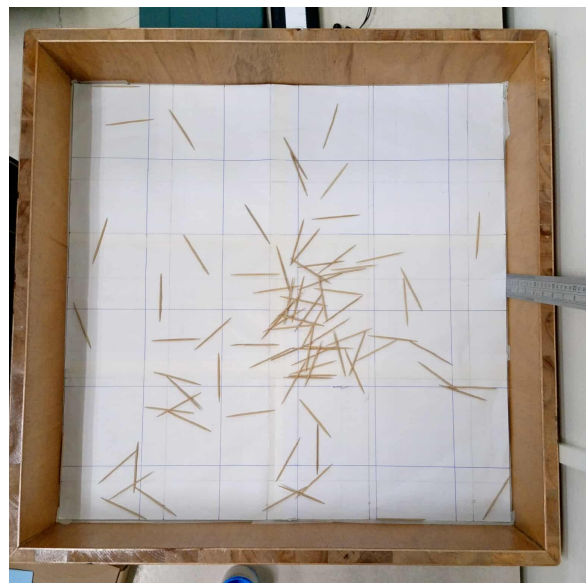
Table 2: Table of Contributors

Member	Contribution
Danielle Louis Dumas	Performed the experiment, wrote Methodology
Vercil Juan	Performed the experiment, wrote Introduction and Results and Discussion
Rwen Josh Narvarte	Performed the experiment, wrote the Abstract and Conclusion

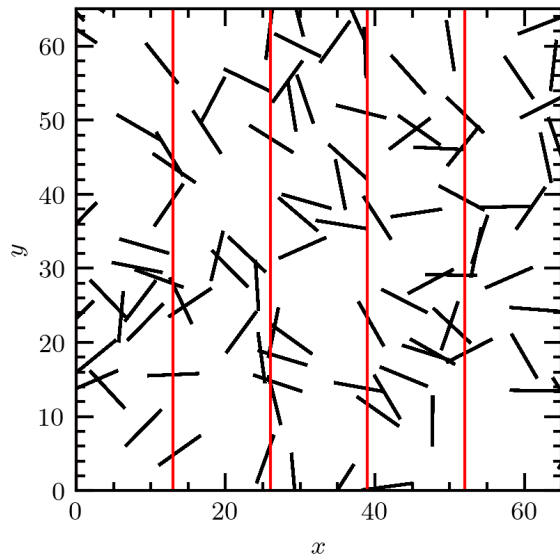
Documentation



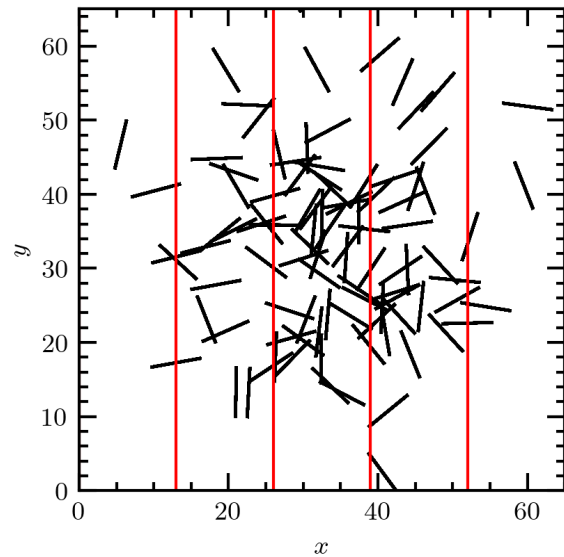
(a) Board with 10 toothpicks



(b) Board with 60 toothpicks



(c) 100 simulated tooth picks with uniform random distribution, in cm. The red line indicates a



(d) 100 simulated tooth picks with gaussian random distribution, in cm. This is more accurate with the physical experiment

Figure 3: Physical and numerical experiments

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Code

All the code used for the graphing and statistical analysis for this report can be found in:

<https://github.com/silvernuke911/Physics157/tree/main/PHYSICS192%20CODE/E5B>